Lesson notes · for you

A connected world: unit check

IoT · Level 4 · Unit check

Time
45 min

Materials
Paper and a pen. No device needed.

Learning goals

- How things talk
- Pass it on: mesh networks
- Keep connected things safe
- Your connected project

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Worksheet

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Name _____ Class _____ Date _____

1 **Twenty soil sensors on a farm 4 km from school run on batteries, and each sends a few bytes an hour. The school can't pay for monthly data plans. Which network suits them best?**

◯ Wi-Fi ◯ LoRa ◯ A mobile network

2 **A sensor that only uses 2G can stop working when its country switches off 2G, even though nothing on the sensor has broken.**

◯ True ◯ False

3 **Match each word to its meaning.**

Hop	• • • •	Doing the harmless thing when a part breaks
Relay	• • • •	Hijacked devices controlled together
Gateway	• • • •	Passes a network's messages to the internet
Botnet	• • • •	A radio that only passes messages on
Fail safe	• • • •	One step of a message from one device to the next

4 **Find the weak point in a line of sensors.**

Sensors stand 200 m apart along a river, and each radio reaches 250 m. A message from the sensor 1,000 m from the gateway needs _____ hops. If the sensor nearest the gateway breaks, every sensor beyond it is _____.

Word bank: 5 · cut off · 4 · rerouted

5 **Which would have kept Mirai out of a camera?**

☐ Cleaning the lens ☐ Changing its default password ☐ Covering the lens with tape

☐ Switching off remote access it didn't need ☐ Buying a camera with a sharper picture